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Design Project Part I

Schematic Designs:

Our group has decided to design a static ripple-carry adder. The figures below will show the creation and functionality of our adder circuit. Standard sizing rules were followed for the devices, with a minimum width length for any transistor of 150nm. The size of each transistor is shown on the schematic. A bit pitch of 2.91um was chosen to make the routing connection easy, avoiding DRC violations.
After a full adder has been constructed, we extend this to the project parameters of two 8 bit values with a control (subtract) function. This is handled by an additional input “subtract” that is fed into every bit level with input B into an XOR gate. Outputs for our full adder are sum and carry. Finally an overflow bit is registered at the output that is the result of the MSB carry and MSB - 1 carry bit fed into another XOR gate.

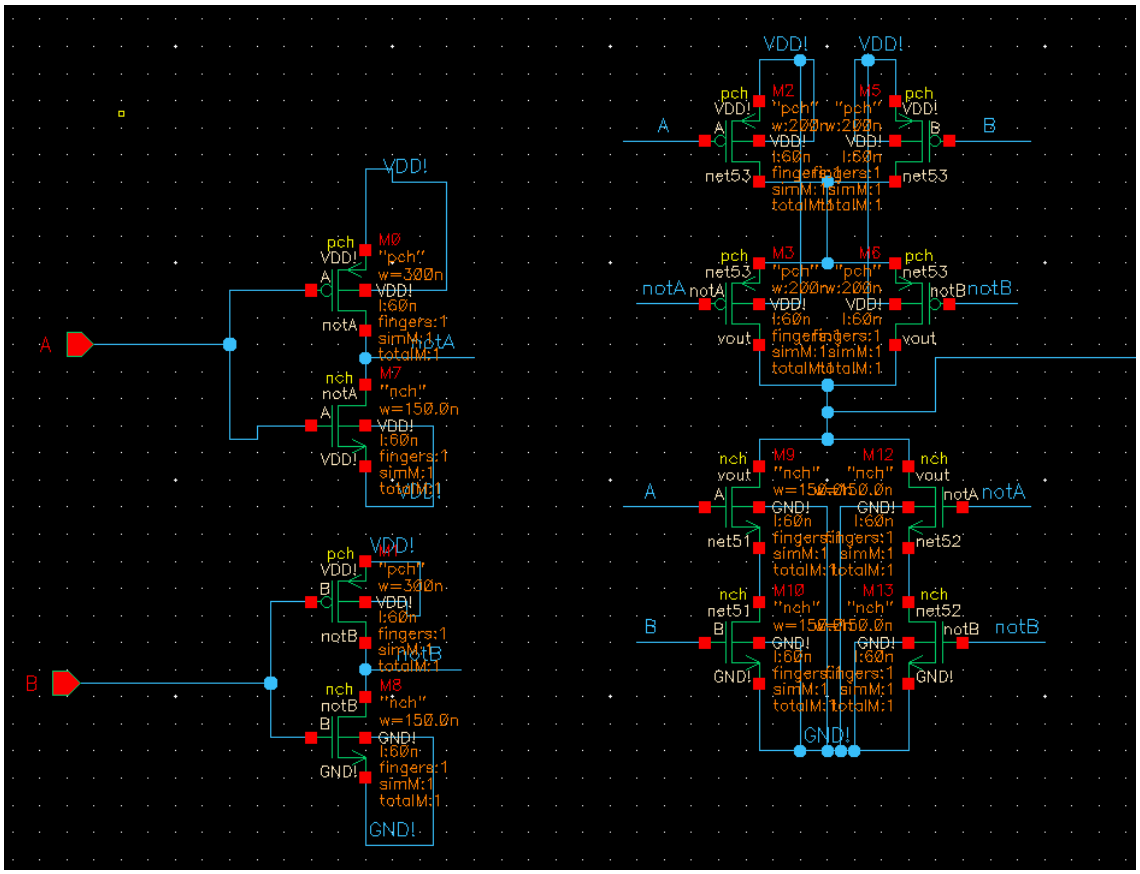


Figure 1: XOR Gate with input A, B and output Vout

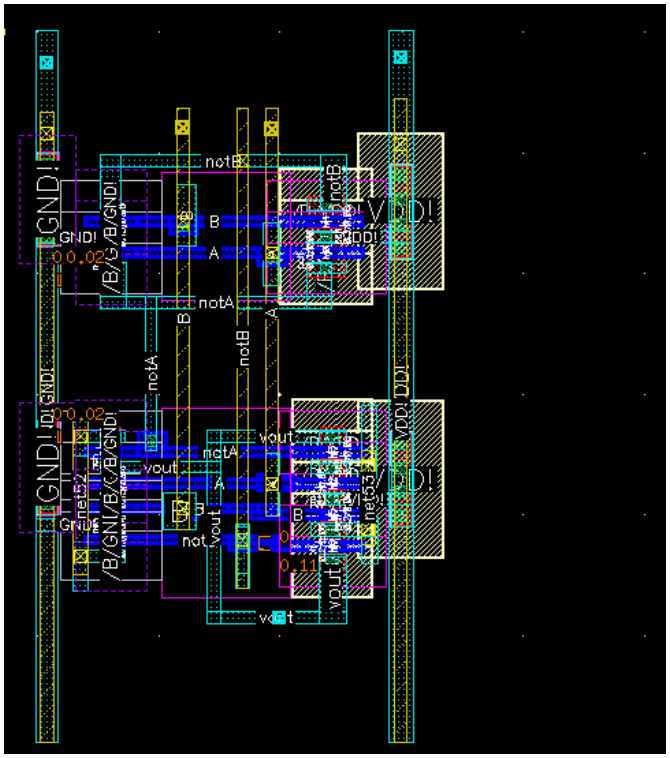
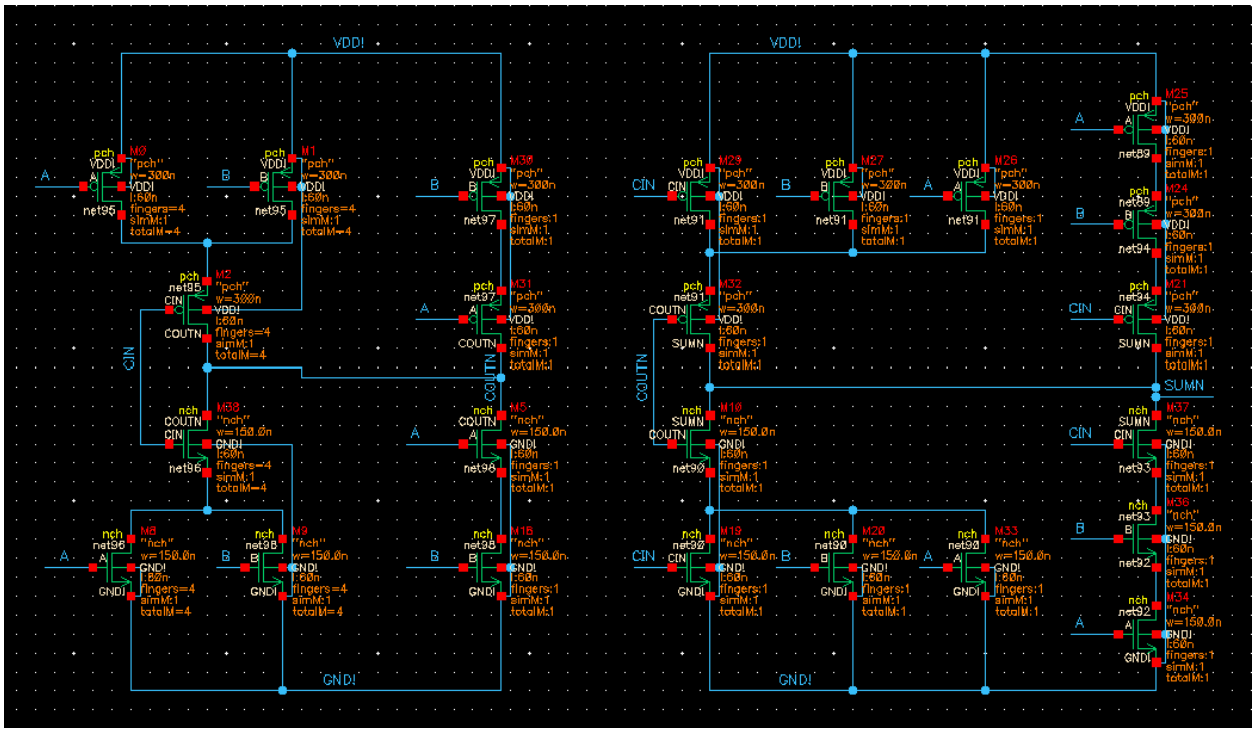


Figure 2: Layout of the XOR Gate shown in Figure 1 with bit pitch of 2.91um.



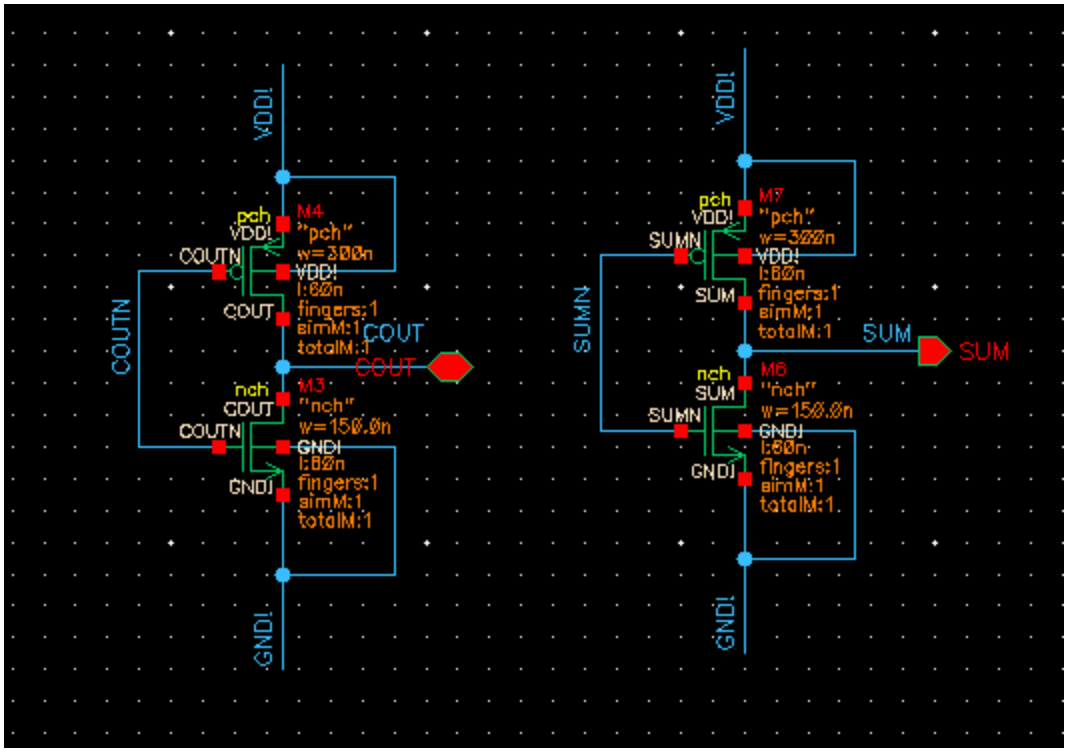


Figure 3: a) Full adder Schematic with input A, B and CIN as well as output SUMN and COUTN. b) Inverters with input SUMN, COUTN and output SUM and COUT.

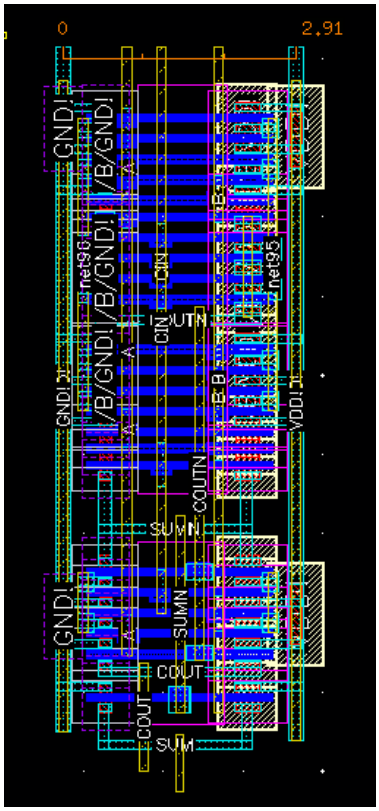


Figure 4: Layout of one full adder with output inverters and bit pitch 2.91um.

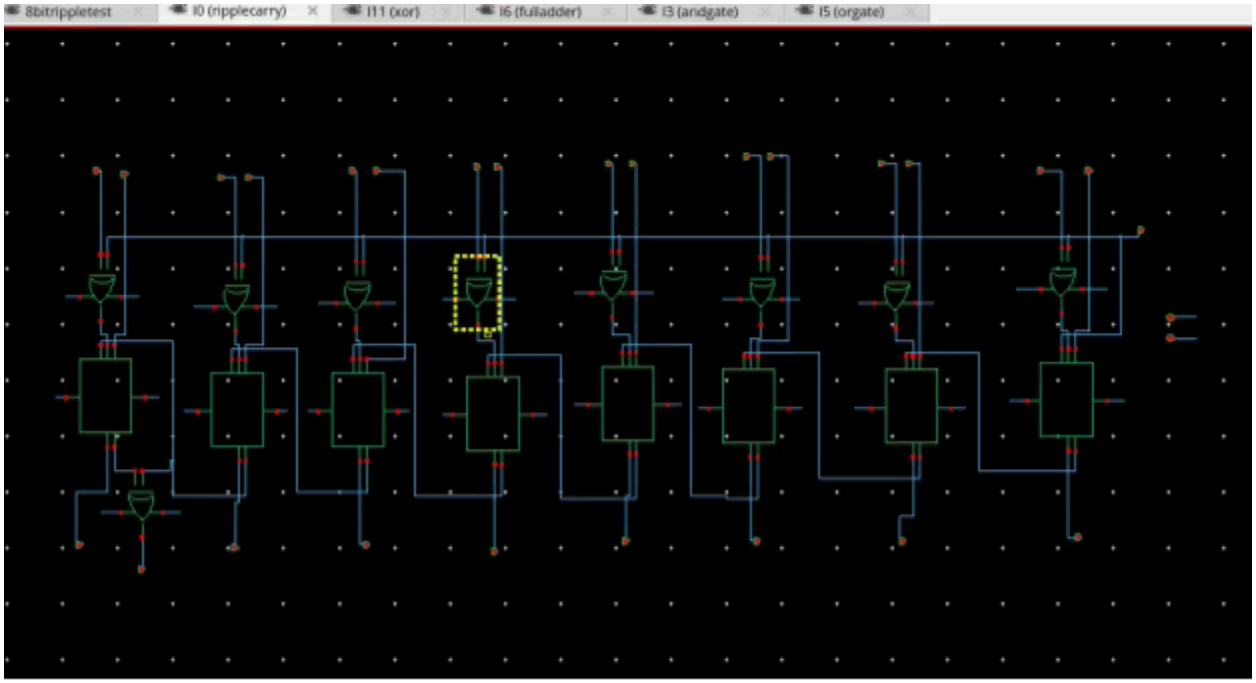
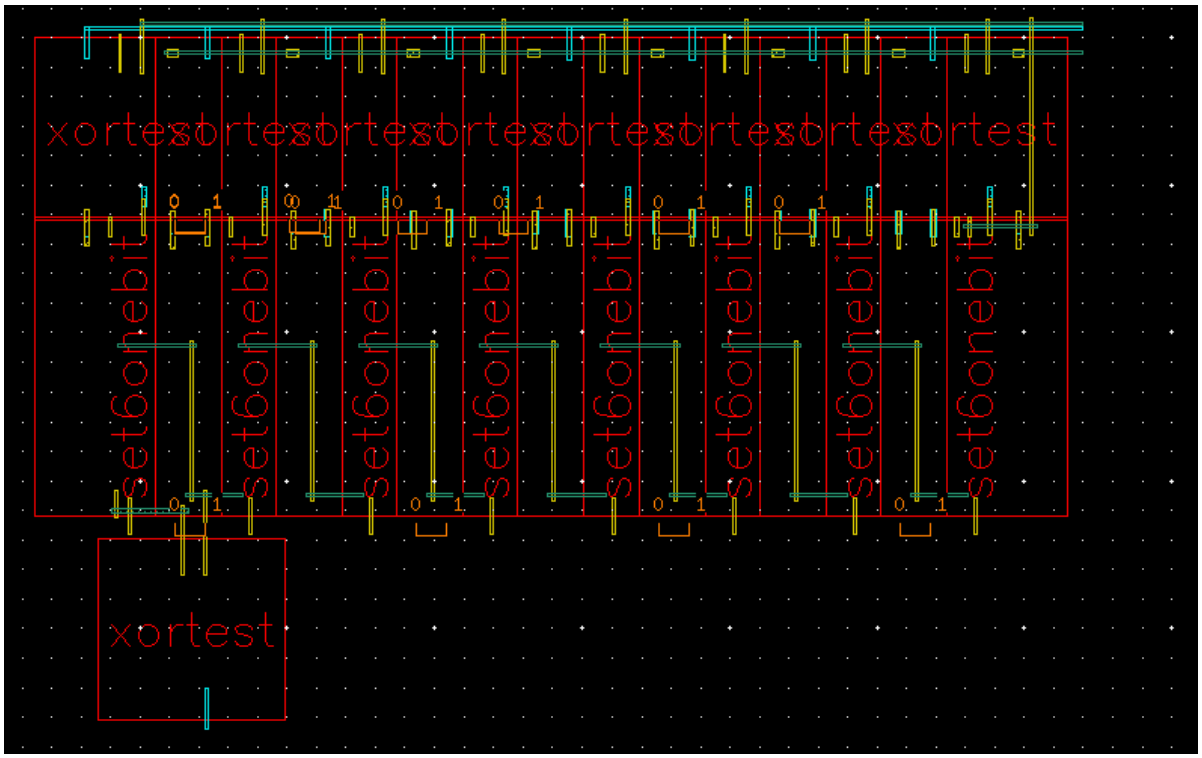


Figure 5: 8-bit Ripple carry adder with outputs Sum<7:0> and 1-bit overflow



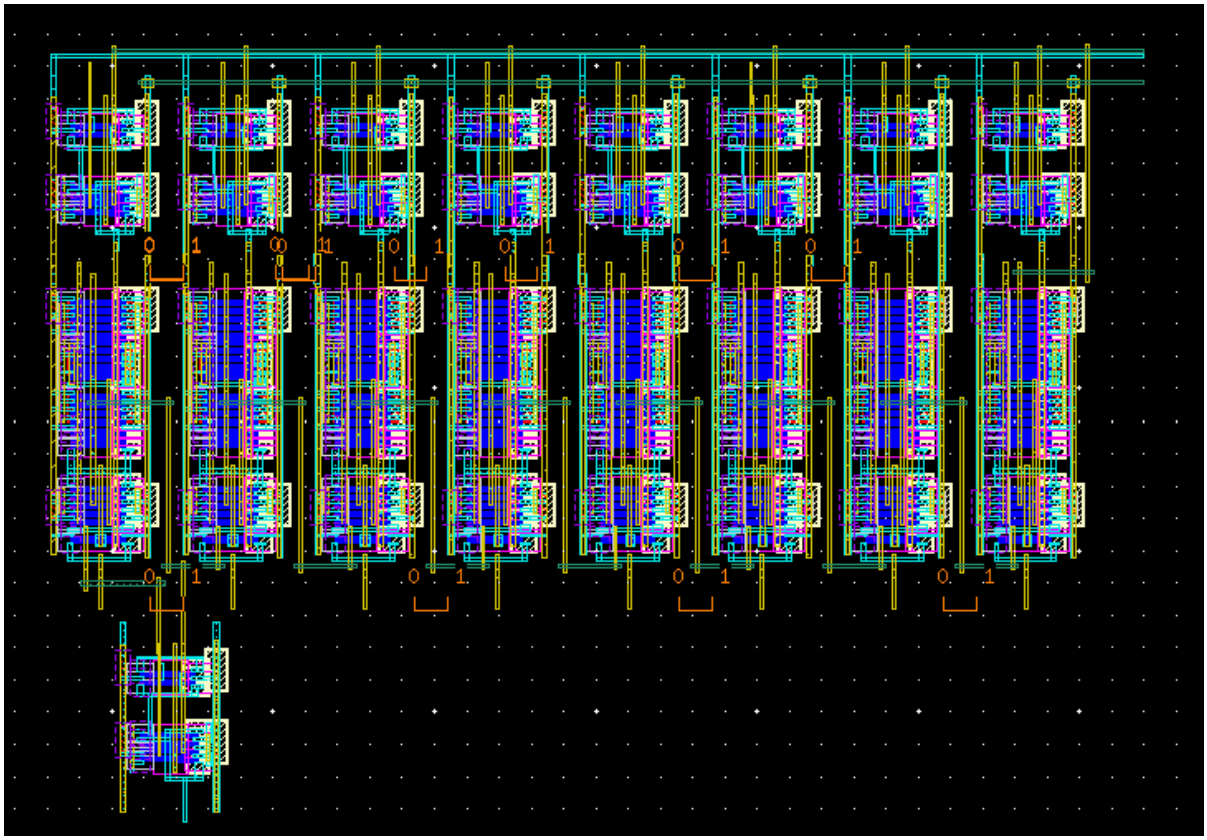


Figure 5: Two views of the final layout with 8 full adders and 9 XOR gates.

Testbench Results

This portion of the design was verifying functionality and critical path delay. Critical path delay is determined by the time of the longest combinational logic path our data needs in order to computate. Our design was verified using a pulse voltage source from analoglib and generating a .txt file to provide random inputs for each bit in our adder and verifying our output. As this is a signed adder the most significant bit (MSB) is our sign bit and the remaining 7 are used for the integer values. Our range is [-128, 127]. Our adder was verified using a variety of numbers to guarantee correctness, but the one example we will provide here is -1(A) + 1(B) = 0. The reason for this example is because we want to use these results for our critical path delay timing. Our critical path delay can be found with any example where we have our carry bit propagate all the way through from our LSB to our MSB.

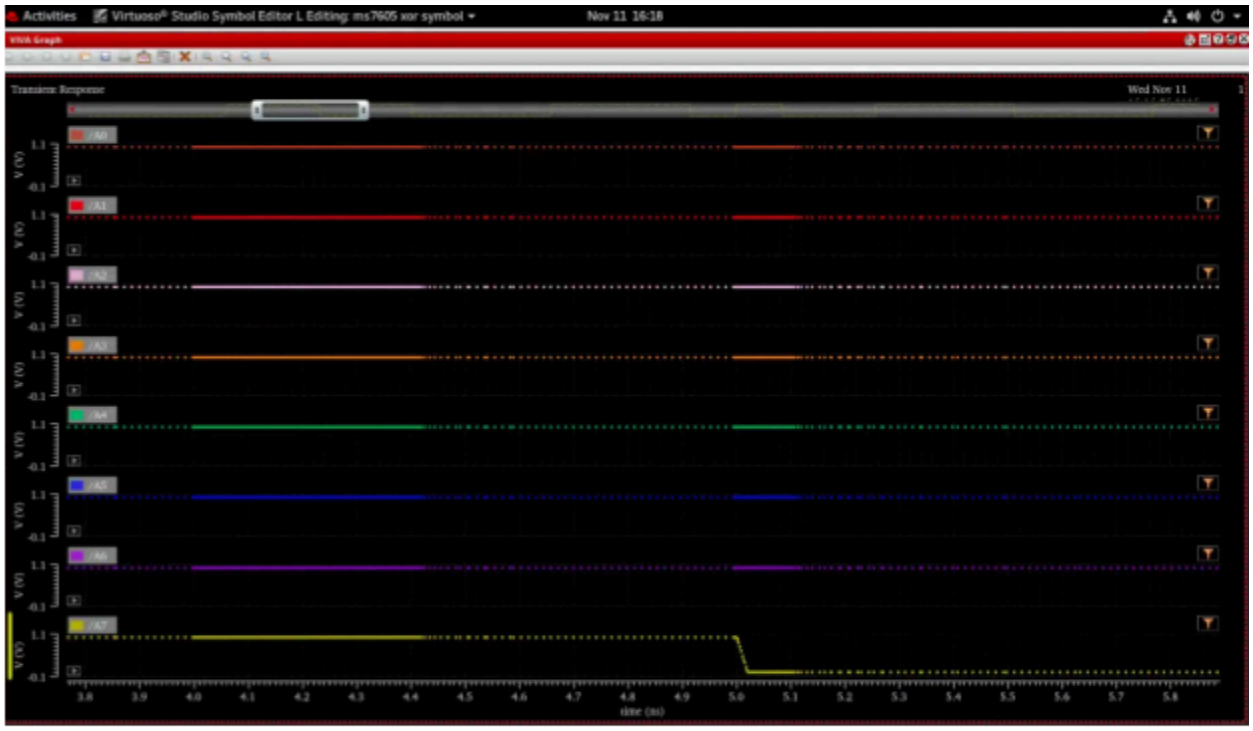


Figure 6: A = 11111111. Signed representation of -1

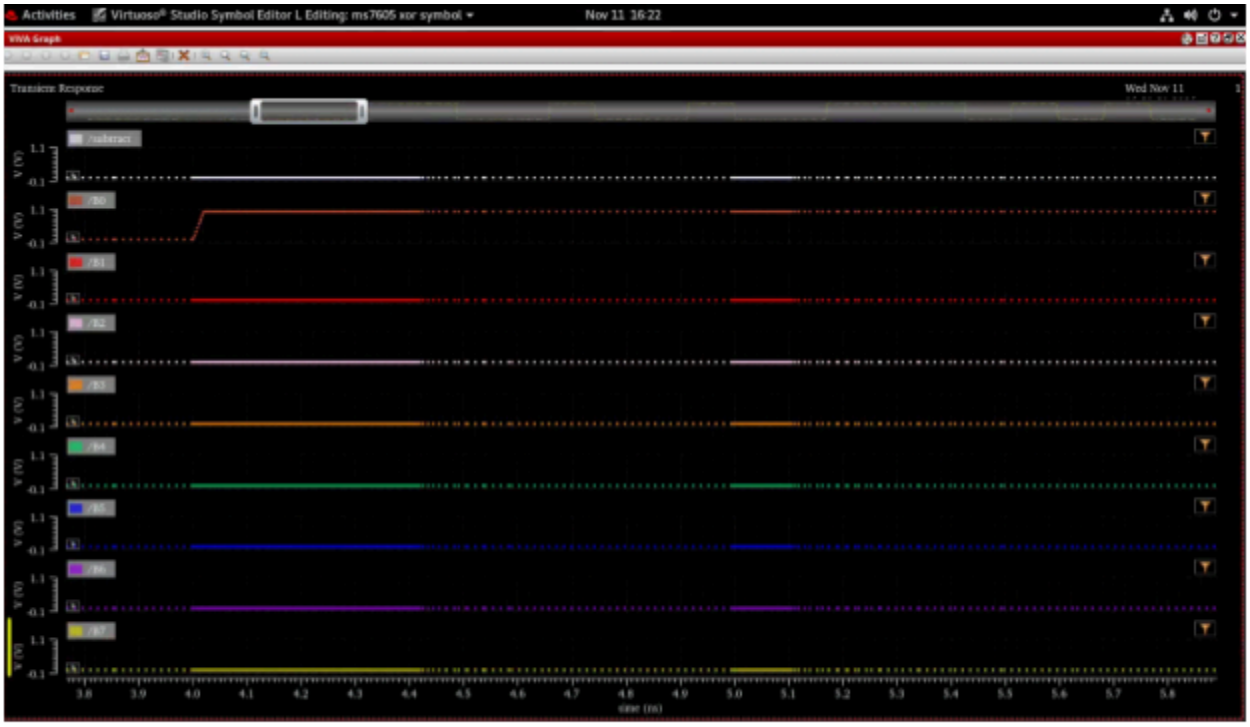


Figure 7: B = 00000001. Signed representation of 1 and subtract is low

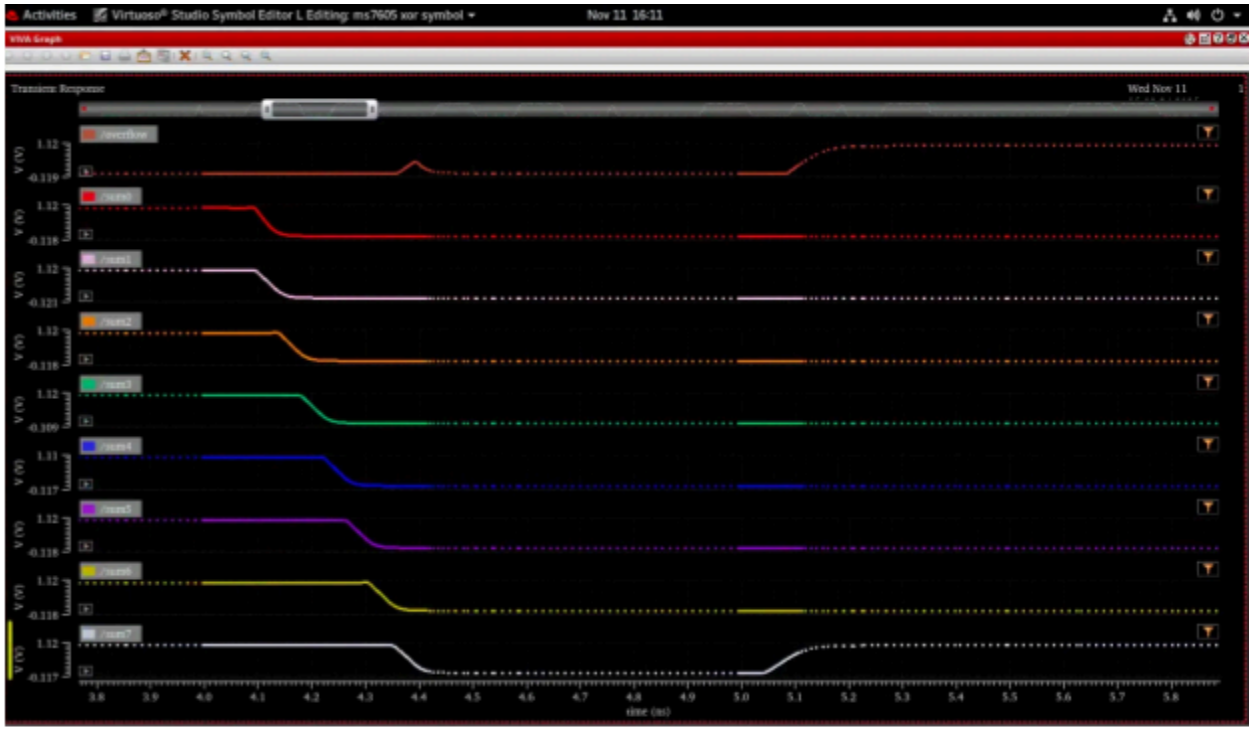


Figure 8: Sum = 00000000 and overflow = 0.

Figures 6-8 also allows us to get a true measurement of our critical path delay. The inputs of A and B change at 4.0ns as shown in Figures 5-6. Figure 7 shows how long it takes for every Sum to update with the bits closer to B0 obviously flipping first. Sum7 takes the longest and crosses over the 50% Voltage threshold after 4.38 ns. B0 has a rise time of 20ps so that crosses over at 4.01 ns making our critical path delay:

4.38 ns - 4.01 ns = 370 ps

For additional verification we can take a glimpse into the next time interval where we update inputs again at 5ns. We change the sign bit from 1 -> 0 making our number A change from -1 to 127. B Stays the exact same as signed integer +1. At around 5.15ns in Figure 8, our overflow bit flips high showing this particular case can't be represented with our 8 bits.